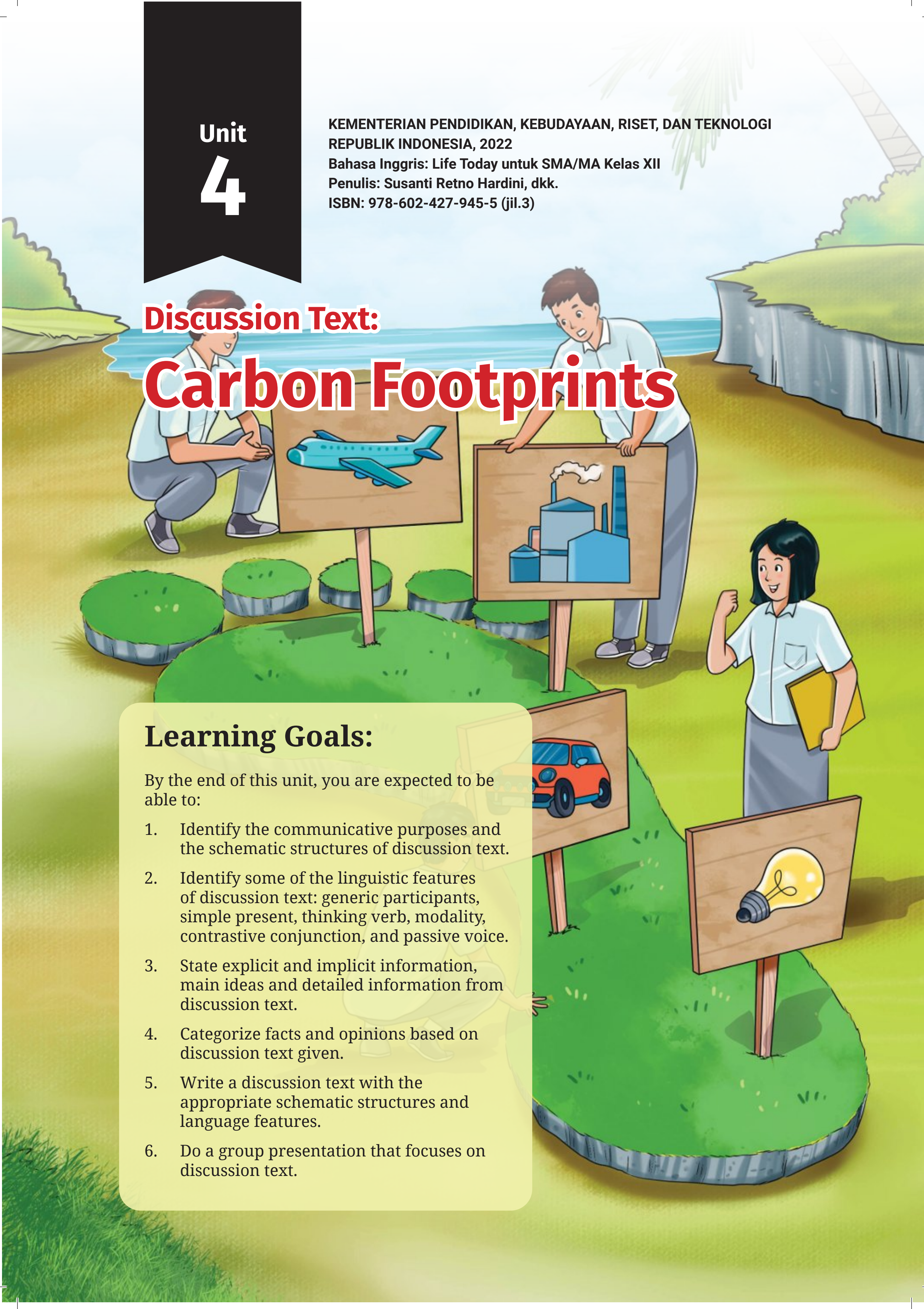

Unit 4

KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI
REPUBLIK INDONESIA, 2022
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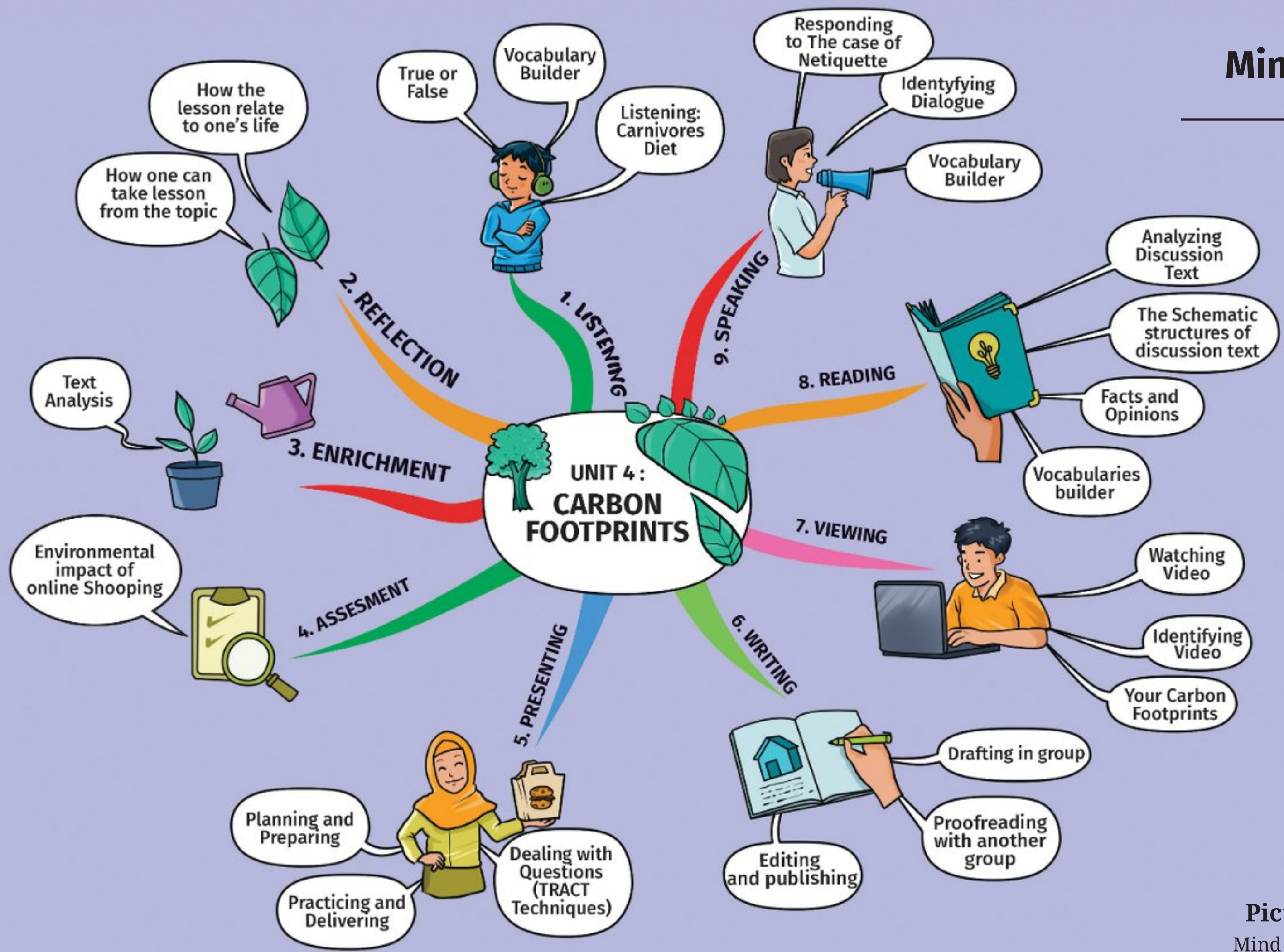
Discussion Text: Carbon Footprints

Learning Goals:

By the end of this unit, you are expected to be able to:

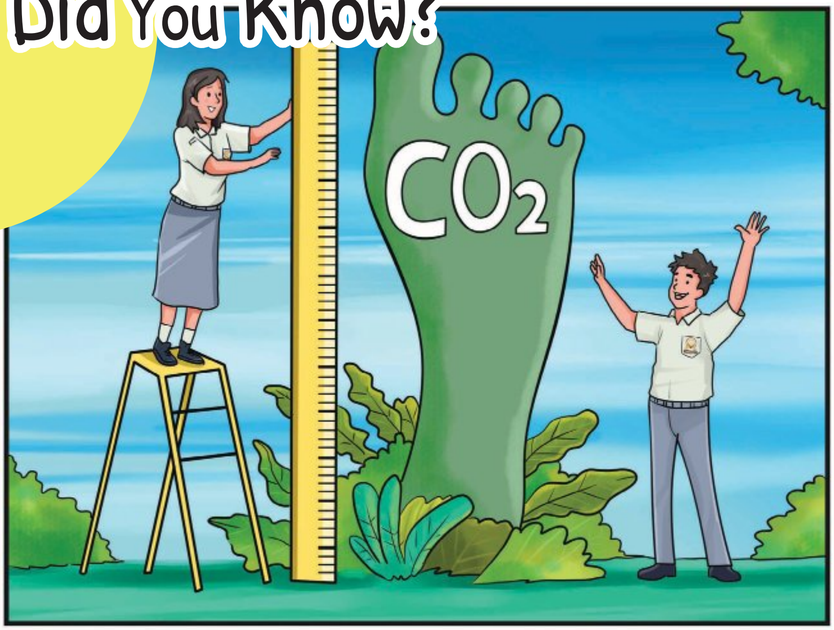
1. Identify the communicative purposes and the schematic structures of discussion text.
2. Identify some of the linguistic features of discussion text: generic participants, simple present, thinking verb, modality, contrastive conjunction, and passive voice.
3. State explicit and implicit information, main ideas and detailed information from discussion text.
4. Categorize facts and opinions based on discussion text given.
5. Write a discussion text with the appropriate schematic structures and language features.
6. Do a group presentation that focuses on discussion text.

Mind Map



Picture 4.1
Mind Map Unit 4

Did You Know?



Picture 4.2 Students calculated their carbon footprint

According to Mike Berners-Lee, a professor at Lancaster University in the UK and author of "The Carbon Footprint of Everything", carbon footprint is the sum total of all the greenhouse gas emissions that had to take place in order for a product to be produced or for an activity to take place, such as: household energy use, transport, food, and everything products we buy, from utensils, clothes, cars to television sets.

Each of these activities and products has its own footprint. For example, a person who regularly consumes beef will have a larger food footprint than his vegan neighbor, but that neighbor's overall footprint may be larger if she drives an hour to work and back in an SUV each day while our meat-eater bicycles to his office nearby. Both their footprints may pale in comparison to the businesswoman across the street, who flies first-class cross-country twice a month.

Then, what about yours?

Adapted from: *https://www.nationalgeographic.com/environment/article/what-is-a-carbon-footprint-how-to-measure-yours*

Listening

Activity 1

Look at the pictures.

Picture 4.3 A female student booked an online means of transportation

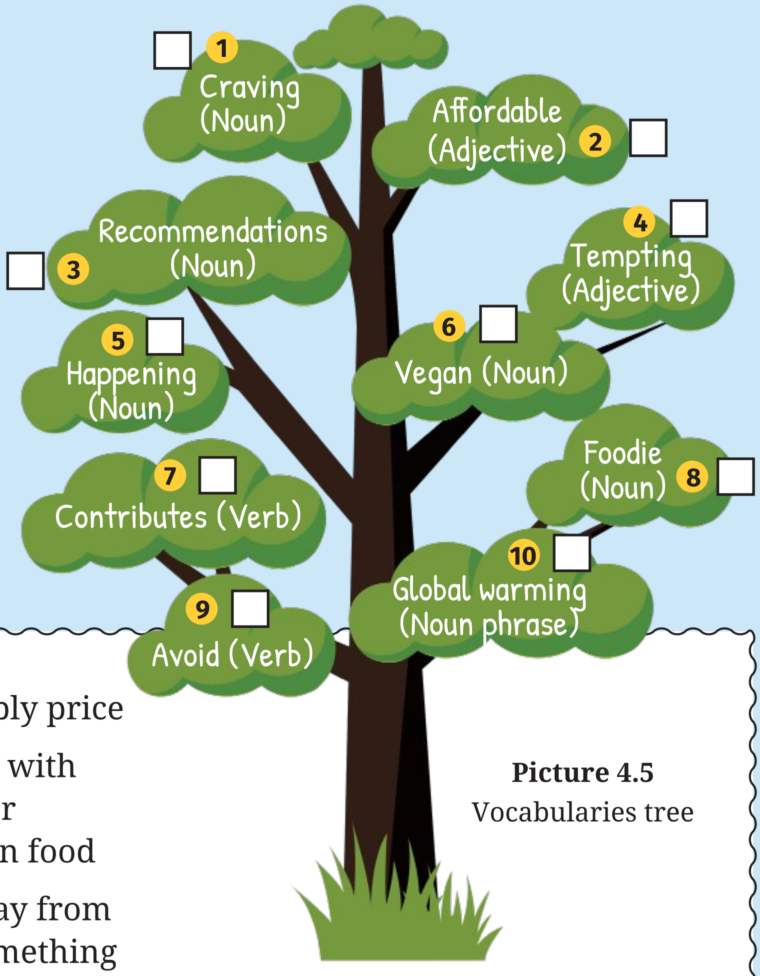
Picture 4.4 Bright home

Discuss the questions in pairs.

1. Are the illustrations above familiar to you?
2. If you were that student, would you do the same?

Activity 2

Work in pairs and match the words/phrases with the correct meanings.



Picture 4.5
Vocabularies tree

a. Reasonably price

b. A person with particular interest in food

c. Keep away from doing something

d. A gradual increase in the overall temperature of the earth's atmosphere is generally attributed to the greenhouse effect caused by increased levels of carbon dioxide, chlorofluorocarbons, and other pollutants.

e. A person who does not eat any food derived from animals and who typically does not use other animal products.

f. Give in order to help achieve or provide something.

g. Powerful desire for something

h. Appealing to

i. An event or occurrence

j. Suggestion to the best action

Activity 3

Listen to the dialogue between two students; Putra and Siti. Siti was in class during break time and saw Putra busy scrolling down his phone.

Based on the dialogue, decide if the statements below are **true** (T) or **false** (F) or **not mentioned** (NM). Put a tick (✓) in the column provided.

NO	STATEMENT	T	F	NM
1	Putra is a foodie.		✓	
2	Putra’s sister is a vegan.			
3	Siti is a vegan herself.			
4	Putra is interested in finding a steak restaurant.			
5	Putra decides to go to the Dinosaur.			
6	Eating meat is the number one contributor to the heat we have been experiencing lately: global warming.			
7	Putra will book at the Dinner Table.			
8	The Dinner Table serves vegan steak.			
9	Putra is a superhero of the earth today.			
10	Public figures influence their followers' behavior a lot.			

Activity 4

Put a tick in the suitable column whether Yes or No whether each action below relates to reducing climate change and carbon footprints.

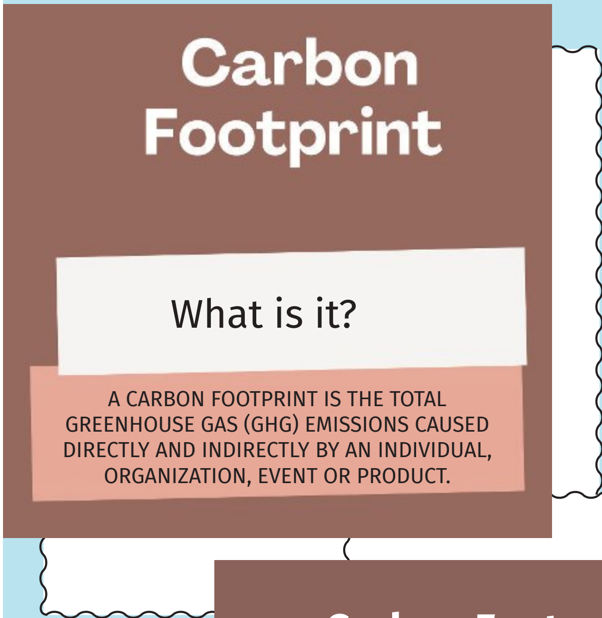
No	Actions	Yes	No
1.	Switch car to public transport	✓	
2.	Recycle waste comprehensively		
3.	Switch to canvas bags		
4.	Vote for extra plastic bags		
5.	Buy only local food		
6.	Waste no extra food		
7.	Use electric cycle		
8.	Wash laundry in cold water		
9.	Don't litter		
10.	Eat vegan diet		
11.	Live car free		
12.	Upgrade fashion interest		
13.	Buy only unpackaged food		
14.	Upgrade gadget		
15.	Upgrade to motorcycle hybrid		

Adapted *from: https://theconversation.com/carbon-footprints-are-hard-to-understand-heres-what-you-need-to-know-144317*

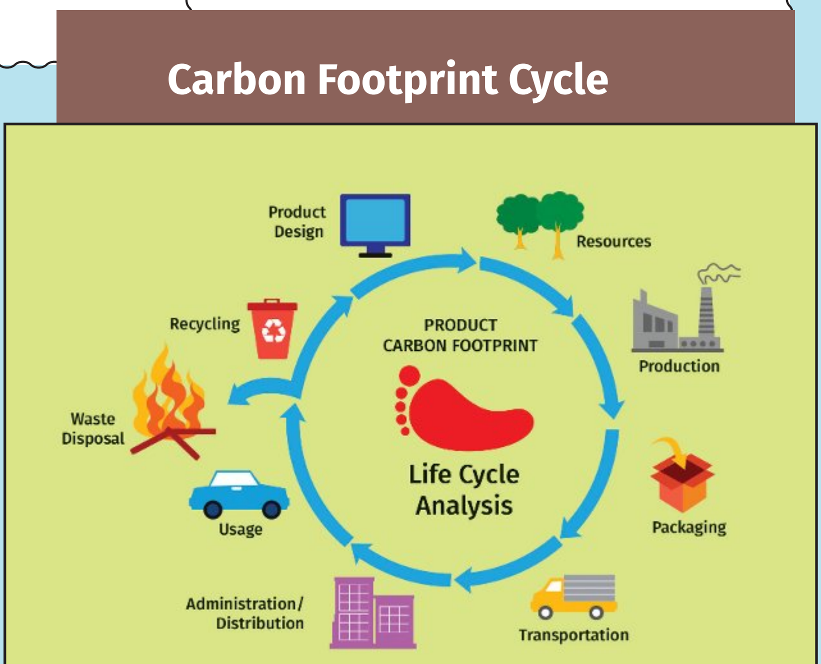
Speaking

Activity 1

Study and discuss the following poster in groups of 4. Pay attention to the meaning of carbon footprint and its cycle.



Picture 4.6
Carbon footprint infographic



Picture 4.7 Carbon footprint cycle

Activity 2

Look at the words/phrases in the left column. Discuss their meanings in the right column in your group.

Car-sharing	Some people travel together to save fuel energy.
Recycling	Used items to produce something useful.
Energy efficiency	Using less energy for the same function.
Vegan diet	Avoid all animal products including meat, egg and dairy.
Traditional energy sources	Things used to create energy with environmental impact
Food waste	Food we don't eat
Emission	Poisonous gasses produced by cars and other vehicles

Home working

Working from home to reduce environmental impact.

Adapted *from: https://www.teachingenglish.org.uk/sites/teacheng/files/Online_class_presentation_Family_Footprint.pdf*

Activity 3

Work in pairs. Classify the following words into four categories of carbon footprint. You may choose one appropriate category among transportation, waste, food and energy. Number 1 has been done for you as an example.

No.	Words/phrases	Category
1	car-sharing	transportation
2	recycling	
3	energy efficiency	
4	vegan diet	
5	traditional energy sources	
6	food waste	
7	emission	
8	home working	

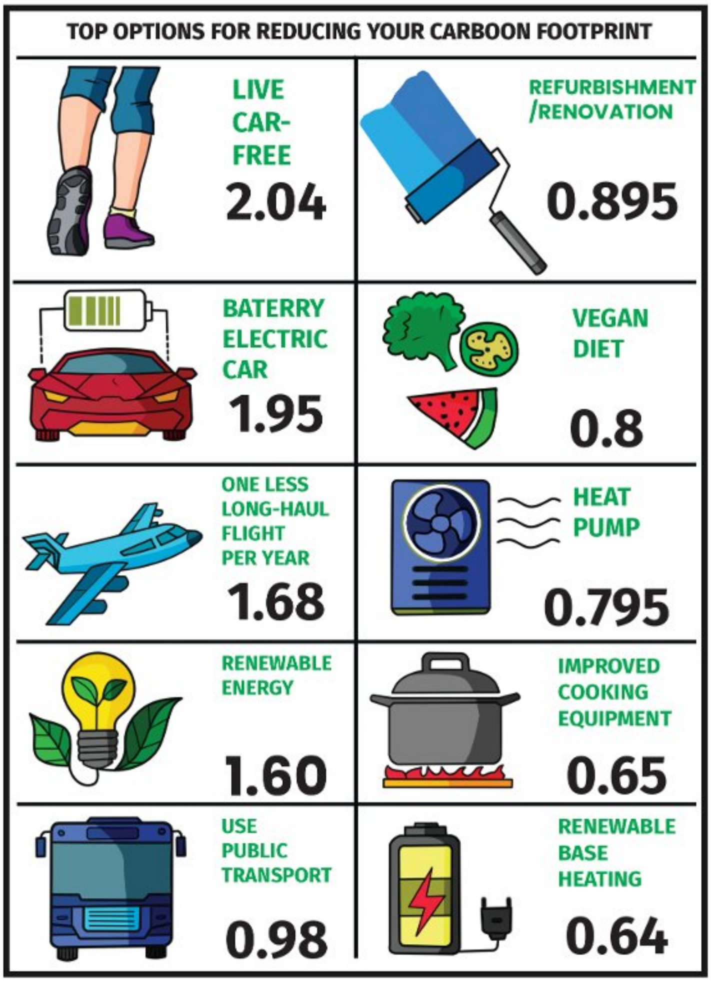
Activity 4

Study the following chart containing several actions to reduce carbon footprint in pairs. State your opinion completed with the data. Do it orally in turn.

This is an example for you:

Student A : In my opinion, conducting a vegan diet won't contribute much in reducing carbon footprint.

Student B : But, at least, it reduces 0.8 tonnes CO2 per year.



Picture 4.8 Options to reduce carbon footprint

Activity 5

Work in a group of 4. Interview your friend to think at least two more actions on how to limit your carbon footprint by completing the table.

This is an example for you:

Student A : What do you think we should do to reduce food dealing with the increasing number of carbon footprint in our country?

Student B : I think we can increase the consumption of healthy and natural food.

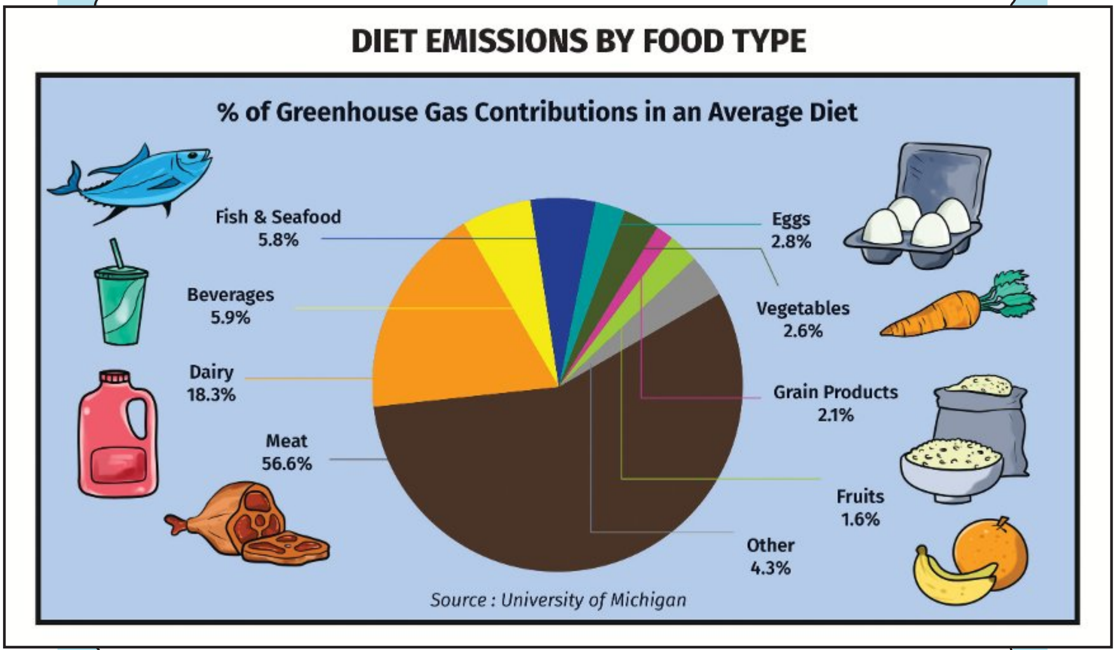
<table>
<tr><th>No.</th><th>Component</th><th>Actions</th><th>Your friends' name</th></tr>
<tr><td rowspan="4">1</td><td rowspan="4">Food</td><td>Increase the consumption of healthy and natural food.</td><td rowspan="4"></td></tr>
<tr><td>Consume local and seasonal products.</td></tr>
<tr><td>....................................
....
....................................</td></tr>
<tr><td>....................................
....
....................................</td></tr>
</table>

2	Clothing	Buy responsibly-made clothes (made from recycled or eco-friendly materials)	
		Take good care of your clothes	
		....	
		....	
3	Transportation	Use public transportation	
		Do carpool	
		....	
		....	
4	Energy and waste	Don’t leave your battery still charging when it is already full.	
		Turn off the water when you brush your teeth.	
		....	
		....	

Reading

Activity 1

A. Observe the picture.



Picture 4.9 Diet emission by food type

B. Answer the questions orally.

1. What does the picture describe?
2. Which one has the most and the least contribution to carbon footprints?

B. Read and learn the vocabulary before you read the text. Consult your dictionary for further information.

Carbon Footprints

(noun phrase)

a measurement of the amount of carbon dioxide produced by the activities of a person, company, organization, etc.

Essential

(adjective)

necessary; needed

Consumption

(noun)

the amount used or eaten

Meat eaters

(noun phrase)

someone who eats meat or has a particular attitude towards meat

Livestock

(noun)

animals and birds that are kept on a farm, such as cows, sheep, or chickens

Greenhouse gasses

(noun phrases)

gasses that causes the greenhouse effect, especially carbon dioxide

Emissions

(noun)

the act of sending out gas, heat, light, etc.

Methane

(noun)

a gas with no smell or color, often used as a fuel

Deforestation

(noun)

the cutting down of trees in a large area, or the destruction of forests by people

Habitable land

(noun phrase)

providing land conditions that are good enough to live in or on

Plant-based foods

(noun phrase)

food consisting or made completely of plants, or mainly of plants

Deficits

(noun)

the total amount by which money spent is more than money received

Nourish

(verb)

to provide people or living things with food in order to make them grow and keep them healthy

Sufficient

(adjective)

enough for a particular purpose

sustainability

(noun)

the quality of being able to continue over a period of time

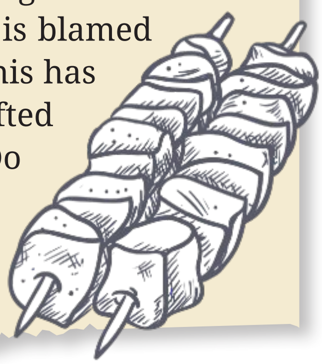
Activity 2

Read the text carefully

Picture 4.10 Meat eaters

Do we need to stop eating meat?

By nature, humans are meat eaters, and our bodies are designed for eating meat. We have incisors for tearing meat, and molars for grinding it. Meat is an essential source of nutrients and calories for a large part of the human population, and this in itself is one major argument for meat-eating. However, meat consumption is blamed for one of the carbon footprints of food. This has caused people to realize and they have shifted from being meat eaters to vegetarians. Do we need to stop eating meat to help protect the planet?



One of the most significant contributors to today's most serious environmental problems is livestock. Globally, the UN estimates it makes up more than 14% of all man-made greenhouse gasses, including methane. The impact of livestock on emissions varies between countries. When we talk about emissions, we usually think of carbon dioxide (CO2). But livestock's emissions also include methane, which is up to 34 times more damaging to the environment over 100 years than CO2, according to the UN. Beef produces the most greenhouse gas emissions, which include methane. At one stage, beef production is the leading cause of deforestation in tropical rainforests which adds to the environmental impact of beef from that part of the world. Moreover, the true climate impact of what we eat is not easy to calculate as carbon footprints of food vary with how it is produced and where it comes from, and thus changes with the seasons. Therefore, how we produce our food doesn't just affect our global emissions, but has a wider environmental impact, such as on biodiversity.

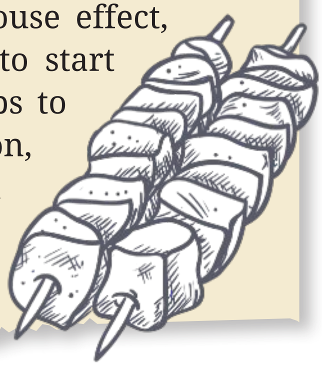
In the report issued by ourworldindata.org, it is said that half of all habitable land is used for agriculture, and three-quarters of that land is used to feed and raise livestock. For this reason, it is assumed that to feed a growing world population, it's far more efficient to use land to produce crops that people can consume directly, and to have a fair global approach ensuring that parts of the world with diets high in meat and dairy shift towards more plant-based foods. Thus, some people choose not to eat meat when other options exist to meet their nutritional needs.

But then on the other hand, meat is an essential source of nutrients and

calories for a large part of the human population, and this in itself is one major argument for meat-eating. Meat is a ready source of protein, vitamin B-12, fat, iron, zinc, and many more essential nutrients that the human body needs to survive. All B vitamins are found in greater concentration in meats than in plant sources, and vitamin B-12 can only be found in animal sources. Besides, the B vitamins are critical to health, especially mental health and deficits in these vitamins can cause confusion, impaired senses, aggression, insomnia, weakness, dementia, and peripheral neuropathy. Moreover, for body builders, eating meat is to get creatine, a nitrogen-containing compound that improves protein synthesis and provides muscles with energy, encouraging muscle gain. In other words, you can take protein supplements, but the best source of protein is meat.

The last argument is that the environmental effects of factory farming have also been criticized, particularly the waste produced during raising and slaughtering and the high cost of grain-based meat production. Fortunately, there are alternatives, such as you can support small farms that raise animals compassionately, follow better farming practices, avoid antibiotics or hormones, and feed animals natural diets.

After looking at all these facts, people can't just stop eating meat for health and environmental reasons. However, since the Earth is suffering due to the greenhouse effect, people should be aware that they need to start to reduce meat consumption since it helps to maintain the Earth's lifespan. For this reason, I am sure that people should start to think and act for the better.



Activity 3

Answer the questions based on the text "Do we need to stop eating meat?"

1. What is the communicative purpose of the text?
   A. To explain the issue and the effect of carbon footprints to the environment.
   B. To persuade people to be vegetarians for the sake of the environment.
   C. To show readers that being vegetarian is the best choice to protect the earth from carbon footprints.
   D. To present two different points of view about the effect of being vegetarians and meat eaters.
   E. To inform the reader that livestock can cause carbon footprints to the environment.
2. The statement which supports the writer's position is ...
   A. People can support small farms that raise animals compassionately by avoiding antibiotics or hormones, and feed animals natural diets.
   B. Deficits in B vitamins can cause confusion, impaired senses, aggression, insomnia, weakness, dementia, and peripheral neuropathy.
   C. Beef produces the most greenhouse gas emissions, which include methane, which also lead to carbon footprints.
   D. Human body needs a source of protein, vitamin B-12, fat, iron, zinc, and many more essential nutrients to survive from meat.
   E. People cannot get protein from supplements as good as from meat.

3. Could the following statements represent why we need to stop eating meat? Tick **Yes** or **No** for each statement.

Could the following statements represent why we need to stop eating meat?	**Yes**	**No**
According to the UN, livestock's emissions are up to 34 times more damaging to the environment over 100 years than CO2.		
The most greenhouse gas emissions, which include methane, are produced by beef.		
The waste produced during raising and slaughtering livestocks and the high cost of grain-based meat production adds to the environmental impact.		
Better farming practices, avoiding antibiotics or hormones, and feeding animals with natural diets are the alternatives which can be done to protect the environment.		
It is far more efficient to feed a growing world population by using land to produce crops that people can consume directly so that parts of the world with diets high in meat and dairy shift towards more plant-based foods.		

4. *Globally, the UN estimates it makes up more than 14% of all man-made greenhouse gasses, including methane.* (Paragraph 2)

   The word 'it' in the sentence refers to ....

   A. The UN
   B. greenhouse gasses
   C. methane
   D. livestock
   E. the impact of livestock

5. According to the text, why do bodybuilders need to eat meat?

   A. To maintain their body shape as bodybuilders.
   B. To provide muscles with energy, supporting muscle gain.
   C. To obtain creatine that improves protein synthesis.
   D. To consume protein supplements is not as good as eating meat.
   E. To fuel your body with the right nutrients.

6. Thinking about the arguments presented by the writer, which argument do you agree with most strongly?

   In your own words, explain your choice by stating your own personal opinion. You can also refer to the arguments presented by the writer.

   ____
____
____
____

7. Some statements are matters of opinion, based on the ideas and values of the writer. Some statements are matters of fact, which may be tested objectively and are either correct or incorrect. Type “matter of opinion” or “matter of fact” next to each of the statements from the writer’s arguments listed below.

   The first one has been done for you.

No.	Arguments from writer’s arguments	opinion or fact
1.	For this reason, I am sure that people should start to think and act for the better.	opinion
2.	So, you have to fuel your body with the right nutrients to achieve sufficient muscle repair and recovery to make gains.	
3.	Fortunately, there are alternatives, such as you can support small farms that raise animals compassionately, follow better farming practices, avoid antibiotics or hormones, and feed animals natural diets.	
4.	But livestock's emissions also include methane, which is up to 34 times more damaging to the environment over 100 years than CO2, according to the UN.	

5.	All B vitamins are found in greater concentration in meats than in plant sources, and vitamin B-12 can only be found in animal sources.	
6.	For this reason, it is assumed that to feed a growing world population, it's far more efficient to use land to produce crops that people can consume directly, and to have a fair global approach ensuring that parts of the world with diets high in meat and dairy shift towards more plant-based foods.	

8. The main idea of paragraph 2 is ....
   A. Livestock is one of the most significant contributors to today’s most serious environmental problems.
   B. Beef produces the most greenhouse gas emissions, which include methane.
   C. Beef production is the leading cause of deforestation in tropical rainforests.
   D. Some people choose not to eat meat because they find other options to meet their nutritional needs.
   E. How we produce our food does not only affect our global emissions but also has a wider environmental impact, such as on biodiversity.

9. What can we conclude from paragraph 5?
   A. The major argument of meat eating is that meat is an essential source of nutrients and calories for a large part of the human population.
   B. People can only find vitamin B-12 in animal sources rather than in plant sources.
   C. It is suggested that people should follow better farming practices, avoid antibiotics or hormones, and feed animals natural diets to protect the environment.
   D. Eating meat is crucial for bodybuilders to get creatine, a nitrogen-containing compound that improves protein synthesis and provides muscles with energy to encourage muscle gain.
   E. Meat is a ready source of protein, vitamin B-12, fat, iron, zinc, and many more essential nutrients that the human body needs to survive.

Activity 4

Now it's time to explore the communicative purpose, the schematic structures and the language features of discussion text.

Communicative purpose of discussion text

Communicative purpose of discussion text	To present arguments and information from different points of view.

Schematic Structures of discussion text

Schematic Structures	Function
Statement of Issue	Tells the reader the problem and what will be argued about it. Gives information about the issue and how it is to be framed.
Arguments for	Tells the reader points to be developed
Arguments against	Tells the reader points to be developed
Recommendation	Tells the reader the position held by the writer. It is also presented at the most logical conclusion. It recommends a final position on the issue

Language Features of discussion text

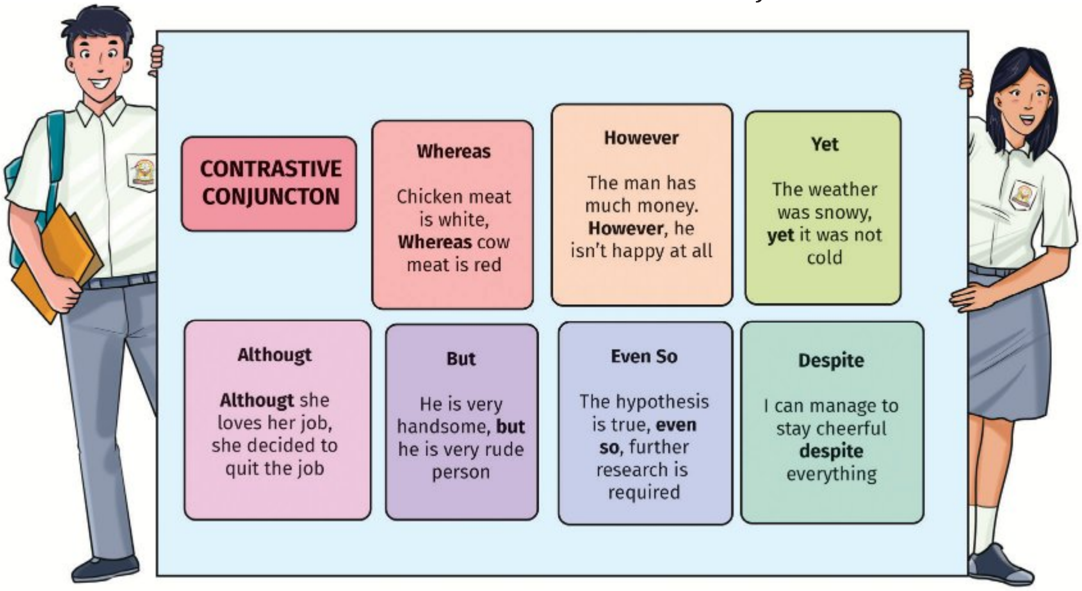
Language Features	Phrases
Generic Participants	carbon footprints; meat
Simple Present	"...,humans are meat eaters,..."

Thinking verb	“I believe....”
Modality	“...,we cannot protect ...”
Contrastive Conjunction	“But then on the other hand, meat is an essential source of nutrients and calories for a large part of the human population, and this in itself is one major argument for meat-eating.”
Passive voice	“All B vitamins are found in greater concentration in meats than in plant sources, and vitamin B-12 can only be found in animal sources.”

Learn the examples of language features:

1. Generic participants: sun; rain.
2. Simple present: “Beef produces the most greenhouse gas emissions, which include methane.”
3. Thinking verbs: think; believe: realize; decide; hope; forget; conclude.
4. Modality: must; should; could; may.
5. Contrastive conjunction: but; on the other hand; however; on other side; although.
6. Passive voice: is produced; is said; is assumed; is used.

Read and learn more about Contrastive Conjunction



Picture 4.11 Contrastive conjunctions

Read and learn more about **Passive Voice**.

Passive voice is used in English when **the thing done** is more important than **the doer** or when the doer is generally known.

Active and Passive Voice		
Tense	**Active**	**Passive**
Present simple	Reporters **write** news reports.	News reports **are written** by reporters.
Present continuous	Tata is **baking** a brownie.	A brownie **is being baked** by Tata.
Past simple	The company **hired** new workers last year.	New workers **were hired** by the company last year.

Past continuous	The salesman **was helping** the customer when the thief came into the store.	The customer **was being helped** by the salesman when the thief came into the store.
Present perfect	They **have already discussed** the book.	The book **has already been discussed**.
Past perfect	He **had delivered** the letters.	The letters **had been delivered**.
Auxiliary	**Active**	**Passive**
Future simple	The company **will hire** new workers.	New workers **will be hired** by the company.
Infinitive	She **has to deliver** the letters.	The letters **have to be delivered**.
Modals	She **must deliver** the letters.	The letters **must be delivered**.

Activity 5

Read again and learn the schematic structures and language features in the text entitled "Do We Need to Stop Eating Meat?"

Structure	Text	Language features
Title	**Do We Need to Stop Eating Meat?**	
Statement of Issue	By nature, humans **are** meat eaters, and our bodies are designed for eating **meat**. We have incisors for tearing meat, and molars for grinding it. Meat is an essential source of nutrients and calories for a large part of the human population, and this in itself is one major argument for meat-eating. However, meat consumption is blamed for one of the carbon footprints of food. This has caused people to realize and they have shifted from being meat eaters to vegetarians. Do we need to stop eating meat to help protect the planet?	are= **simple present** meat= **generic participant**

Arguments for:	One of the most significant contributors to today's most serious environmental problems is livestock. Globally, the UN estimates it makes up more than 14% of all man-made greenhouse gasses, including methane. The impact of livestock on emissions varies between countries. When we talk about emissions, we usually think of carbon dioxide (CO2). But livestock's emissions also include methane, which is up to 34 times more damaging to the environment over 100 years than CO2, according to the UN. Beef produces the most greenhouse gas emissions, which include methane. At one stage, beef production is the leading cause of deforestation in tropical rainforests which adds to the environmental impact of beef from that part of the world. Moreover, the true climate impact of what we eat is not easy to calculate as carbon footprints of food vary with how it is produced and where it comes from, and thus changes with the seasons.	

	Therefore, how we produce our food doesn't just affect our global emissions, but has a wider environmental impact, such as on biodiversity. In the report issued by ourworldindata.org, it is said that half of all habitable land is used for agriculture, and three-quarters of that land is used to feed and raise livestock. For this reason, **it is assumed** that to feed a growing world population, it's far more efficient to use land to produce crops that people can consume directly, and to have a fair global approach ensuring that parts of the world with diets high in meat and dairy shift towards more plant-based foods. Thus, some people choose not to eat meat when other options exist to meet their nutritional needs.	... is assumed...= passive voice
Arguments against:	**But then on the other hand**, meat is an essential source of nutrients and calories for a large part of the human population, and this in itself	But = **contrastive conjunction**

is one major argument for meat-eating. Meat is a ready source of protein, vitamin B-12, fat, iron, zinc, and many more essential nutrients that the human body needs to survive. All B vitamins are found in greater concentration in meats than in plant sources, and vitamin B-12 can only be found in animal sources. Besides, the B vitamins are critical to health, especially mental health and deficits in these vitamins can cause confusion, impaired senses, aggression, insomnia, weakness, dementia, and peripheral neuropathy. Moreover, for body builders, eating meat is to get creatine, a nitrogen-containing compound that improves protein synthesis and provides muscles with energy, encouraging muscle gain. In other words, you can take protein supplements, but the best source of protein is meat.

The last argument is that the environmental effects of factory farming have also been criticized, particularly the waste produced during

on the other hand= **contrastive conjunction**

	raising and slaughtering and the high cost of grain-based meat production. Fortunately, there are alternatives, such as you can support small farms that raise animals compassionately, follow better farming practices, avoid antibiotics or hormones, and feed animals natural diets.	
Recommen-dation	After looking at all these facts, I **believe** people can't just stop eating meat for health and environmental reasons. However, since the Earth is suffering due to the greenhouse effect, people should be aware that they need to start to reduce meat consumption since it helps to maintain the Earth's lifespan. For this reason, I am sure that people **should** start to think and act for the better.	believe = **thinking verb** should= **modality**

Activity 6

According to the reading text of Activity 5, discuss the questions in a group of 3 or 4.

1. What is the generic participant?
2. What is the issue being discussed?
3. Mention at least two of the writer's arguments that he/she is for the issue?
4. Mention at least two of the writer's arguments that he/she is against the issue?
5. What is the writer's position?Is he for or against the issue? Explain.
6. Identify the passive voice taken from the text by classifying the "thing done" and the "doer". One example has been done for you.
    1. **Passive Voice**: Moreover, the true climate impact of what we eat is not easy to calculate as carbon footprints of food vary with how **it is produced** and where it comes from, and thus changes with the seasons.

       Thing done: food is produced

       Doer: not mentioned/ people (generally known)
    2. **Passive voice**: In the report, **it is said** that half of all habitable land is used for agriculture, ... and three-quarters of **that land is used** to feed and raise livestock.

       Thing done:

       Doer:

3. **Passive voice**: ...and three-quarters of **that land is used** to feed and raise livestock.

   Thing done:

   Doer:

4. **Passive voice**: For this reason, **it is assumed** that to feed a growing world population, ...

   Thing done:

   Doer:

5. **Passive voice**: **All B vitamins are found** in greater concentration in meats than in plant sources, and vitamin B-12 can only be found in animal sources

   Thing done:

   Doer:

Viewing

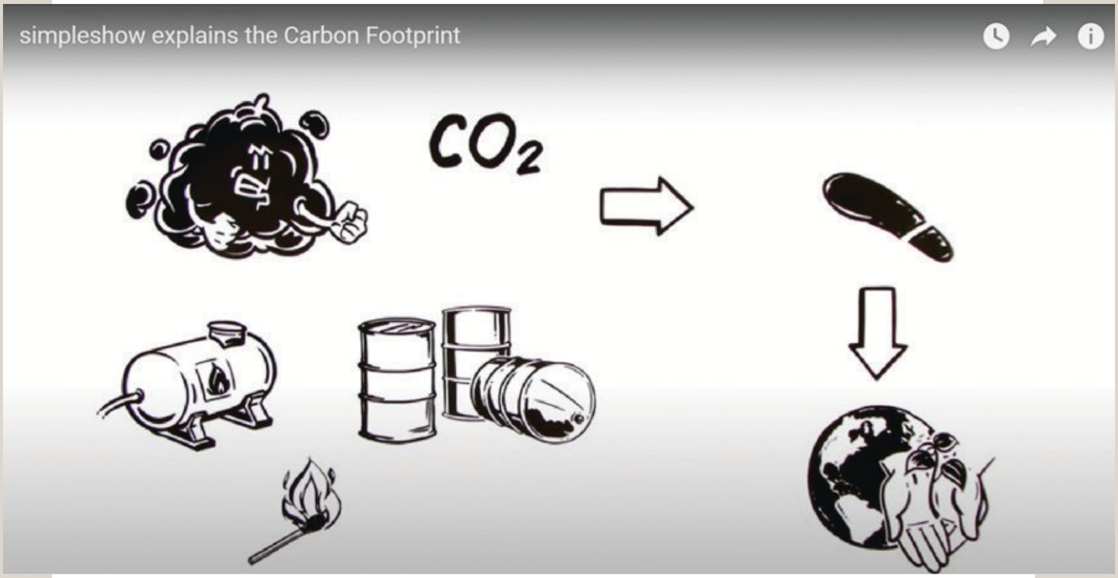
Picture 4.12 Carbon footprint from vehicle and factories

Activity 1

Watch the first video carefully.

The Carbon Footprint

Simpleshow explains the Carbon Footprint



Picture 4.13 Carbon footprint by YouTube/Simpleshow

You've probably been hearing a lot about climate change and how you should reduce your carbon footprint. But what's that exactly? Just like an actual footprint, it's a mark you leave upon the environment. No, not with your shoes but with every action that releases "Carbons". Those are the harmful gasses, such as CO2, which are pumped out by burning fossil fuels, like oil or gas. And the more fuel is used, the bigger your footprint will be. You may think that by driving your car, the only carbons you release come from the engine, but no. Consider the carbons that are emitted just to get fuel into the tank: From the energy needed to extract the oil from underground, the pollution caused by transportation and refinement, to the final delivery to your local petrol station. Not to mention the CO2 released by manufacturing your car in the first place. More than you thought, eh? So, unless you live in a cave; you and everything you own have its own carbon footprint: reading a book -- Printing and distributing it uses energy. Brush your teeth and your tools will have a history in a factory. Even something as basic as an apple could have traveled hundreds or even thousands of miles to end up in your local supermarket. You see, it's pretty much impossible to leave no carbon footprint behind. But by thinking about your actions and personal choices, maybe you can make your feet just that little bit smaller and really help to put the boot into climate change.

Answer these questions based on the video you have watched.

1. What does the video tell you about?
2. What activities contribute to climate change?

Activity 2

Watch the second video.

Can healthy food save the planet?

https://www.youtube.com/watch?v=PIc42oIU0Ik

Picture 4.14 Healthy food

You know the saying "you are what you eat." But the way we currently eat is in fact ruining our health, the health of others and that of the planet. Unhealthier food is now deadlier than alcohol, drug, and tobacco use combined. 2.1 billion people are overweight yet we eat more sugar, fat, and red meat than ever. Still, 821 million go to bed hungry every night. On top of that our food is the main cause behind species extinction and a third of all global greenhouse gas emissions. So, can we feed a growing population without destroying the planet and ourselves?

Science had no clear answer to this question. That's why EAT gathered 37 of the world's best scientists to determine what a healthy and sustainable diet is and how to get there. The result is the EAT-Lancet commission – a scientific blueprint for a healthy and sustainable future.

If we change the way we produce, consume, transport, and waste food, we can feed everyone a healthy diet while improving the health of our planet. What does this look like?

Meat can stay on our planet but plants need to be the new main course. We should eat a huge variety of vegetables and fruits, and a low amount of meat, dairy, and seafood. We should choose unsaturated fats and stay away from refined grains, highly processed and added sugar, and we have no food to waste.

It will take huge changes but following this plan will lower our risk of cancer, strokes, and diabetes. It could help 11 million adult deaths per year. In fact, consuming and producing food more efficiently and mindfully will help to keep our planet flourishing.

We have an answer now: we know the right course for a better future. It is on us to actually take that step. Our food can be the key to solving the biggest challenges we face: food really can fix it.

*EAT is a global, non-profit startup dedicated to transforming our global food system through sound science, impatient disruption and novel partnerships.

Answer these questions based on the video you have watched.

1. What does the video tell you about?
2. What activities contribute to climate change?

Activity 3

After watching the two videos, in pairs, check your understanding by giving a circle/color/checklist on the correct statement as presented (in the video). One number has been answered for you.

No	Which statements are correct?	
1	Every human in the world now contributes to climate change (v1).	Only people in the city contribute to climate change (v1).
2	Carbon footprints can be released by anyone and anything (v.1).	Carbon footprints can be released by humans' activity with machines (v.1).
3	This video disagrees with more impacts of climate change by humans' health (v.2).	This video concerns more impacts of climate change by humans' unhealthy diet (v2).
4	The United Nations made a change for the climate change issue while improving the health of the earth (v.2).	People can make a change for the climate change issue while improving the health of the earth (v.2).

5	The presenter felt if people stayed in the cave that would reduce carbon footprint (v.1).	The presenter asked the viewers to stay in the cave to reduce carbon footprint (v.1).
6	This video talks about the carbon footprint shared by general human activities (v.1).	This video talks about the carbon footprint shared by forest and animal activities (v.1).
7	Printing and distributing a book will surely cause carbon release (v.1).	Printing and distributing a book does not cause carbon release (v.1).
8	The apple you bought in the market will not leave a carbon footprint (v.1).	The apple you bought in the market will, in fact, leave a carbon footprint (v.1).
9	Humans' diet is the main cause behind species extinction and a third of all global greenhouse gas emissions (v.2).	Humans' diet is the third main cause behind species extinction and the main cause of global greenhouse gas emissions (v.2).
10	Carbons are harmful gasses produced merely by fossil fuels (v.1).	Carbons are harmful gasses produced by all human activities (v.1).

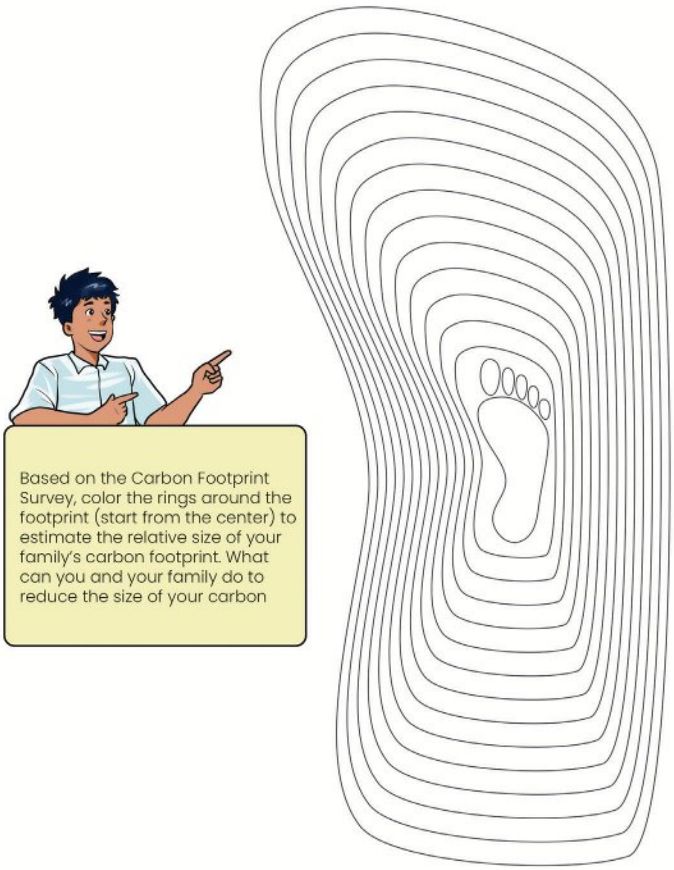
Activity 4 (Optional)

How huge is your carbon footprint?

The Carbon Footprint Survey will ask a series of questions that will direct the participant to color lines around the footprint drawing. The more greenhouse gasses you produce, based on your answers, the bigger the carbon footprint grows. Different color crayons will represent the four categories of behavior surveyed:

Housing and Home Energy	Transportation
1. If you live in a single-family home, color 4 rings **RED**; if you live in an apartment or other type of home, color 2 rings **RED**. 2. If you don't use energy-efficient light bulbs such as CFLs (compact fluorescent), color 1 more ring **RED**. 3. If your home doesn't have a programmable thermostat, color 1 more ring **RED**. 4. If you are not familiar with the Energy Star appliance rating system, color 1 more ring **RED**.	1. For every small car in your family, color 1 ring **BLUE**. 2. For every medium or large car in your family, color 2 rings **BLUE**. 3. If you don't regularly change the air filter on your car and check the tire pressure, color 1 more ring **BLUE**. 4. For every airplane trip you've taken in the past year, color 1 more ring **BLUE**.
	Recycling and Waste
1. If you are a vegetarian, color 1 ring **GREEN**; if you are not a vegetarian, color 2 rings **GREEN**.	If you usually recycle your household trash, color 1 ring **BROWN**; if you never recycle, color 2 rings **BROWN**.

2. If you never eat organic food, color 1 more ring **GREEN**. 3. If you take baths, run the faucet while brushing your teeth or washing dishes, or water your lawn several times a week, color 1 ring **GREEN**.	2. If you never compost your yard and kitchen waste, color 1 more ring **BROWN**.



Picture 4.15 Coloring carbon footprint

Writing

In the previous sections, you have learned all the information about carbon footprints and discussion text. Now it is the time for you to do a writing project.

Activity 1: Plan

Make a group of 2-4 students. Brainstorm ideas before you start making a text. Follow the plan table below for your writing guidance. Consider the punctuation, capitalization and tenses. You can do the task by writing it in the book or paper or by typing it digitally using google docs or other docs apps.

Topic/ Title: 1. What is the topic of your text? 2. What is the possible title for your text?
Statement of issue: 1. What are the problems? 2. Why do you argue about the problem?
Arguments for 1. What are your arguments which support the problem?
Arguments against 1. What are your arguments which disagree with the supporting arguments?
Recommendation 1. What is your position towards the problem? 2. What is your recommendation about the problem?

Activity 2: Writing Plan

In your groups, write the ideas in activity 1 into paragraphs in the table below. Consider the use of contrastive conjunctions, thinking verbs and modality for linking or contrasting ideas between and within paragraphs. You can do the task by writing it in the book or paper or by typing it digitally using google docs or other docs apps.

Topic/ Title:	
Statement of Issue	
Arguments for	
Arguments against	
Recommendation	

Activity 3: Review Text

After writing the discussion text based on the narrative schematic structures, swap the text with another group. Proofread the text by checklisting (✓) the elements of schematic structures and linguistic features found in the text.

Schematic Structures	**Yes/No**	**Linguistic Features**	**Yes/No**
Title		Generic Participants	
Statement of issue		Simple Present	
Arguments for		Thinking verb	

Arguments against		Modality	
Recommendation		Contrastive Conjunction	
		Passive voice	

Activity 4: Redraft Text

Rewrite the text which has been reviewed by another group in the box below. Then submit your final text to your teacher.

Final writing draft

Presenting

In groups, present your discussion text in the form of digital or printed in front of the class.

Consider the following steps:

1. Planning presentation (Asking yourself about 5W+1H questions)
2. Preparing Presentation (Personal notes, Visual, Handout, etc)

3. Practicing Presentation
4. Delivering Presentation
5. Dealing with Questions (TRACT Techniques: Thank the questioner, Repeat the question, Answer the question, Check with the questioner if they are satisfied, Thank them again)

Source: *student-learning.tcd.ie*
https://student-learning.tcd.ie/assets/PDF/Giving_Presentations.pdf

Assessment

Read the following text and answer the questions individually.

Environmental Impact of Online Shopping

In the past few decades, the way we shop has changed dramatically. We used to buy our goods in traditional shops, on the high street or in department stores. Now, customers are increasingly buying online, where they can order whatever they want directly to their door with the click of a mouse. One in seven sales are now made online and studies suggest that by 2021, global online retail will reach an enormous US$4.8 trillion. As companies race to improve their internet shopping experience, the trend towards shopping online is predicted to continue.

But what is the impact of all this online shopping on the environment? You might think that online shopping is greener than in-store shopping. After all, an online store does not use the electricity that a traditional store might use and it doesn't require the customer to drive anywhere. Items are often delivered to several homes at once, so you would think the carbon savings must be significant. Take the typical home delivery round in the UK, for example. Supermarket drivers often do 120 deliveries on an 80-kilometer round, producing 20 kilograms of CO_2 in total. In contrast, a 21-kilometer drive to the store and back for one household would generate 24 times more CO_2!

However, the reality is slightly more complex than that. Many home deliveries fail the first time and the driver has to make a second or third attempt to deliver the purchase. Customers who choose speedy delivery or those who buy single items from different places also contribute towards increasing the carbon footprint.

The carbon footprint also goes up if the customer chooses to return the item. A study in Germany showed that as many as one in three online purchases are returned. According to another study, merchandise worth nearly US$326 million is returned each year in the USA. Two billion kilograms of this ends up in landfill, leading to 13 tonnes of CO_2 being released.

In Indonesia, online returns can involve a number of environmentally damaging activities. Consumers sending items back, and couriers collecting and redistributing them, all means extra driving and thus traffic congestion and carbon emissions. Cleaning, repairing and/or repackaging returned items mean consuming more natural resources and potentially using more materials that contain fossil fuels or palm oils. Processing, transporting and landfill of single-use or non-recyclable packaging used in returns mean more land use and a greater carbon footprint.

Clothing is one product that has high return rates. Unlike in a walk-in store, the online shopper can't try things on before buying. So, companies offer free returns to make it easier for shoppers to purchase the same item of clothing in different sizes and colors. Customers try them at home, keep one and return the rest of them. However, when clothes are returned, they are not always cleaned and put back for sale. This is because many companies have found it cheaper to simply throw away the returned items than to pay someone to sort the damaged goods from the unwanted ones. In these cases, the returned clothes, which might be in perfect condition, end up in landfills or burnt.

When we take all these factors into consideration, we realize that online shopping isn't necessarily as green as people might think. That last kilometer to your door is costly, for companies and for the environment. There is some positive news, as various online retailers are starting to lower their carbon footprint by investing in electric delivery vehicles. However, the question of how to deal with returns efficiently and without waste is a challenge that many companies have not wanted to face. As online shoppers become aware of what companies are doing, and campaign groups demand urgent action in the face of the climate and ecological emergency, there is increasing pressure for companies to take responsibility for the environmental impact of their activities.

1. According to the text, where did we buy our goods? You may choose more than one correct answer.
   A. anywhere
   B. on the high street
   C. in conventional shops
   D. In the department store
   E. online
   F. through catalogue
2. What makes online shopping contribute much to the carbon footprint?
   A. When the customer returns the item he/she bought.
   B. When a person burns his/her goods.
   C. If a smoker keeps smoking while riding the customer.
   D. When customers shop a lot.
   E. If the customer buys many goods for himself/herself.
3. Are the sentences true or false? Put a tick next to each statement.

No	Statements	True	False
A	Online shopping is decreasing in terms of numbers.		
B	More shopping might be done online in the future.		

C	Online shopping uses less electricity than conventional shopping.		
D	The carbon footprint of online shopping is worsened due to speedy delivery and goods returns.		
E	Companies have found environmentally-friendly solutions for the problem of returned goods.		

4. State how is online shopping correlated to the increasing number of carbon footprint.

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Enrichment

Below are some arguments taken from a text. Read the text carefully.

AGREE 1: Switching to renewable energy is not as simple as it is being made out to be. Quite the opposite.	DISAGREE 1: Leaving fossil fuels in the ground is good for everyone
"It is commonly assumed that greenhouse gas and energy problems can be solved by switching from fossil fuel sources of energy to renewables. However, little attention has been given to exploring the limits to renewable energy. Unfortunately, people working on renewable energy technologies tend not to throw critical light on the difficulties and limits. They typically make enthusiastic claims regarding the potential of their specific technologies." (Alex Epstein)	*"To deliver a 50% probability (which is not exactly reassuring) of no more than 2C of warming this century, the world would have to leave two-thirds of its fossil fuel reserves unexploited. I should point out that reserves are just a small fraction of resources (which means all the minerals in the Earth's crust). The reserve is that proportion of a mineral resource which has been discovered, quantified and is viable to exploit in current conditions: in other words that's good to go.... a third of the world's oil reserves, half its gas reserves and 80% of its coal reserves must be left untouched to avert extremely dangerous levels of global warming. 2C is dangerous enough; at present we are on course for around 5C by the time the century ends, with no obvious end in*

"It would be difficult to find a more taken for granted, unquestioned assumption than that it will be possible to substitute renewable energy sources for fossil fuels, while consumer-capitalist society continues on its merry pursuit of limitless affluence and growth. There is a strong case that this assumption is seriously mistaken." (Ted Trainer)	*sight beyond 2100."* (George Monbiot, The Guardian, 2015) *"The major thrust of climate-change claims is that man is destroying the planet. There is much evidence to show that we are the greatest burden that Earth has to bear. To simply rape the earth of all its fossil-fuels would be gross folly."* (Dr. Peter Langdon)
AGREE 2: Renewables cannot provide the required amount of energy to supply demand (Intermittency) *"As you look at the jagged and woefully insufficient bursts of electricity from solar and wind, remember this: some reliable source of energy needed to do the heavy lifting. In the case of Germany, much of that energy is coal. As Germany has paid tens of billions of dollars to subsidize solar panels and windmills, fossil fuel capacity, especially coal, has not been shut down—it has increased.*	**DISAGREE 2: Renewable energy can meet energy needs in a safe and reliable way** *"...The key is to have a mix of sources spread over a wide area: solar and wind power, biogas, biomass and geothermal sources. In the future, ocean energy can contribute. Intelligent technologies can track and manage energy use patterns, provide flexible power that follows demand through the day, use better storage options and group producers together to form virtual power plants.*

Why? Because Germans need more energy, and they cannot rely on the renewables." (Alex Epstein) *"It is concluded that although the foregoing figures are not precise or confident, their magnitudes indicate that it will not be possible to meet a 1000 EJ/yr energy target for 2050 from alternative energy sources, within safe greenhouse gas emission levels… . Such a goal could not be achieved without radical change in social, economic, political and cultural systems."* (Ted Trainer)	*With all these solutions we can secure the renewable energy future needed. We just need smart grids to put it all together and effectively 'keep the lights on'".* (Greenpeace.org 2014) *"There's no shortage of renewable energy from the sun, wind and water and even stuff usually thought of as garbage — dead trees, tree branches, yard clippings, left-over crops, sawdust, even livestock manure, can produce electricity and fuels — resources collectively called 'biomass'… The sunlight … in one day contains more than twice the energy we consume in an entire year. … Clean energy sources can be harnessed to produce electricity, process heat, fuel and valuable chemicals with less impact on the environment."* (California Energy Commission 2006)

Question 1

Based on the text, which of the following issues do people seem to be debating?

A. Renewable energy is replaceable

B. Fossil fuels leave carbon footprints

C. Renewable energy cannot replace fossil fuels

D. Renewable energy needs time to be used

E. Renewable energy can reduce carbon footprints

Question 2

Whom do the arguments belong to? Put a checklist in the table provided

Arguments	Alex	Ted	George	Peter
Without fundamental changes to the social, economic, political, and cultural structures, this objective could not be attained.				
"It is widely assumed that switching from fossil fuel to renewable energy sources will solve the greenhouse gas and energy problems."				

The constraints of renewable energy, however, have not received much attention.				
"The central claim of climate-change denial is that man is destroying the planet."				
"It would be difficult to find a more taken for granted, unquestioned assumption than that renewable energy sources can be substituted for fossil fuels while consumer-capitalist society continues on its merry way of limitless affluence and growth."				
To avoid extremely dangerous levels of global warming, one-third of the world's oil reserves, half of its gas reserves, and 80 percent of its coal reserves must be left untouched.				

There is ample evidence that we are the most significant burden that the Earth must bear. It would be a huge mistake to simply rob the Earth of all its fossil fuels."				

Question 3

Could the following arguments represent Greenpeace's disagreement? Type **Yes** or **No** for **each** argument.

Arguments	**Yes**	**No**
"As you observe the jagged and woefully insufficient bursts of electricity from solar and wind, keep in mind that some reliable source of energy is required to do the heavy lifting."		
Ocean energy has the potential to contribute in the future.		
With all of these solutions, we can ensure the necessary renewable energy future.		
Clean energy sources can be used to generate electricity, process heat, fuel, and valuable chemicals while minimizing environmental impact.		
We simply need smart grids to connect everything and effectively "keep the lights on."		

Question 4

Thinking about the arguments presented by the six sources from the text above, which source do you agree with most strongly?

Source's name: ________________________

Using your own words, explain your choice by referring to your own opinion and the arguments presented by the source.

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Question 5

Based on the text of agreement and disagreement, are the statements in the table below facts or opinions? Put a checklist in the **Fact** or **Opinion** for each statement.

Is the argument a fact or an opinion?	Fact	Opinion
People working on renewable energy technologies typically make enthusiastic claims regarding the potential of their specific technologies."		

Intelligent technologies can track and manage energy use patterns, provide flexible power that follows demand through the day, use better storage options and group producers together to form virtual power plants.		
The reserve is that proportion of a mineral resource which has been discovered, quantified and is viable to exploit in current conditions.		
There is a strong case that this assumption is seriously mistaken.		
However, little attention has been given to exploring the limits to renewable energy.		

Reflection

After studying Unit 4, answer the following reflective questions. Write your answers in the box.

1. Why is it important for us to understand our carbon footprint?
2. How can we help the global warming issue after learning about carbon footprint?

MY REFLECTION OF UNIT 4
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Scan the following QR Code or visit the link to download the audio recording:

https://static.buku.kemdikbud.go.id/content/media/rar/BIng_12.zip

Glossary

A

Abundant (adj):

existing or occuring in large amounts

Adequate (adj):

sufficient for a specific need or requirement.

Affordable (adj):

reasonably price

Apologize (v):

to say sorry for something you've done

Avoid (v):

keep away from doing something

B

Bank account (np):

an account with a bank created by the deposit of money or its equivalent and subject to withdrawal of money (as by check or passbook)

Bill (n) :

a statement of charges for food or drink, goods sold, services performed, or work done

Biomass resources(n):

an organic matter that is available on a renewable or recurring basis

Blackout (n):

a period of darkness caused by a failure of electrical power

C

Carbon dioxide (n):

a heavy colorless gas CO2 that does not support combustion, dissolves in water to form carbonic acid, is formed especially in animal respiration and in the decay or combustion of animal and vegetable matter, is absorbed from the air by plants in photosynthesis, and is used in the carbonation of beverages.

Carbon footprint (n): the amount of carbon dioxide and other carbon compounds emitted due to the consumption of fossil fuels by a particular person, group, etc.

Carbon Footprints (np):

the amount of greenhouse gasses and specifically carbon dioxide emitted by something (such as a person's activities or a product's manufacture and transport) during a given period

Car-sharing (np):

some people travel together to save fuel energy

Cash (n):

money or its equivalent (such as a check) paid for goods or services at the time of purchase or delivery

Cashless (adj):

the exchange of funds by check, debit or credit card, or various electronic methods rather than the use of cash.

Characters (n):

one of the persons of a drama or novel

Charity (n):

the voluntary giving of help, typically in the form of money, to those in need.

Climate change (n):

a change in global or regional climate patterns, in particular a change apparent from the mid to late 20th century onwards and attributed largely to the increased levels of atmospheric carbon dioxide produced by the use of fossil fuels.

Coda (n):

a concluding part of a literary or dramatic work

Conserve (V):

to protect (something, especially an environmentally or culturally important place or thing) from harm or destruction

Contributes (v):

give in order to help achieve or provide something

Craving (n):

powerful desire for something

Credit card (np):

a card authorizing purchases on credit

Currency (n):

a system of money in general use in a particular country.

Cybercrime (n):

criminal activity (such as fraud, theft, or distribution of child pornography) committed using a computer especially to illegally access, transmit, or manipulate data.

D

Deal with something (v):

to take action to solve a problem

Deception (n):

lying; trickery; untruth.

Deduct (v):

subtract or take away (an amount or part) from a total

Deforestation (n):

the action or process of clearing of forests; the state of having been cleared of forests

Devastating (adj):

causing great damage or harm.

Digital (adj):

of, relating to, or utilizing devices constructed or working by the methods or principles of electronic

E

E-commerce platforms (np):

the buying and selling of goods and services, or the transmitting of funds or data, over an electronic network

Emissions (n):

substances discharged into the air (as by a smokestack or an automobile engine)

Encounter (v):

to engage in conflict with

Energy efficiency (np):

using less energy for the same function

E-payments (n):

electronic payments for transactions made on the Internet.

E-wallet (n):

a type of electronic card which is used for transactions made online through a computer or a smartphone

Exposure (n):

the condition of being presented to view or made known.

F

Fanfare (n):

demonstration; parade; protest.

Feel guilty (v):

to feel that something you did was wrong

Flinch (v):

make a quick, nervous movement as an instinctive reaction to fear, pain, or surprise

Flourish (v):

to be in a state of activity or production.

Food waste (n):

food we don't eat

Foodie (n):

a person with particular interest in food

Fraudster (n):

swindler; deceiver; scammer.

Fumbling (n):

doing something awkwardly, especially when using your hands

G

Gas emission (n):

direct emissions produced by burning fuel for power or heat, through chemical reactions, and from leaks from industrial processes or equipment.

Global warming (np):

a gradual increase in the overall temperature of the earth's atmosphere is generally attributed to the greenhouse effect caused by increased levels of carbon dioxide, chlorofluorocarbons, and other pollutants.

Governing force (np):

to exercise authority over; rule, administer, direct, control, manage; to influence the action or conduct of; guide; sway.

Green energy (n):

any energy type that is generated from natural resources, such as; sunlight, wind or water

Greenhouse gasses (np):

any of various gaseous compounds (such as carbon dioxide or methane) that absorb infrared radiation, trap heat in the atmosphere, and contribute to the greenhouse effect

Guarantee (v):

to provide a formal assurance or promise, especially that certain conditions shall be fulfilled relating to a product, service, or transaction.

H

Happening (n):

an event or occurrence

Hate speech (n):

speech expressing hatred of a particular group of people

Hoax (v):

to trick into believing or accepting as a genuine something false and often preposterous

Home working (np):

working from home to reduce environmental impact

Hurt somebody's feeling (v):

to upset somebody by saying bad things about them

Hybrid (n):

something (such as a power plant, vehicle, or electronic circuit) that has two different types of components performing essentially the same function

I

Impersonation (n):

roleplay; acting; imitation.

In person (np):

actually being with someone, rather than communicating online

Indispensable (adj):

absolutely necessary; essential.

Influencer (n) :

a person who inspires or guides the actions of others.

In-joke (np):

a funny story that only certain people understand

Interest (n):

money paid regularly at a particular rate for the use of money lent, or for delaying the repayment of a debt.

K

Kerosene (n):

a flammable hydrocarbon oil usually obtained by distillation, of petroleum and used as a fuel, solvent or thinner

L

Language (n):

a systematic means of communicating ideas or feelings by the use of conventionalized signs, sounds, gestures, or marks having understood meanings

Livestock (n):

animals kept or raised for use or pleasure especially : farm animals kept for use and profit

Local power company (n):

a local utility company supplies electricity to buildings connected to the power grid

M

Meat eaters (np):

someone who eats meat or has a particular attitude towards meat

Money Laundering (n):

the concealment of the origins of illegally obtained money, typically by means of transfers involving foreign banks or legitimate businesses

N

Nourish (v):

to nurture; to promote the growth of

O

Overweight (n):

excessive or burdensome weight.

P

Payment (n):

the act of paying

Persecution (n):

the condition of being persecuted, harrases or annoyed

Phenomenon (n):

a rare or significant fact or event.

Phishing (n):

hacking; spamming; attacking.

Piqued (adj):

to arouse anger or resentment in.

Power outage (np):

(a power cut, a power out, a power failure, a power blackout, a power loss, or a blackout) is the loss of the electrical power network supply to an end user.

Proofread (v):

to read and mark corrections in (something, such as a proof)

R

Receipt (n):

a writing acknowledging the receiving of goods or money

Recommendations (n):

suggestion to the best action

Recycling (n):

used items to produce something useful

Refurbishment (v):

to brighten or freshen up

S

Scaling up (v):

to increase the size, amount, or importance of something, usually an organization or process

Scan (v):

to read or mark so as to show metrical structure

Selfish (adj):

a person or an action lacking consideration for others

Shamefully (adv):

in a way that deserves blame

Shun (v):

to avoid something

Siphoned off (v):

to dishonestly take money from an organization or other supply, and use it for a purpose for which it was not intended

Spill the tea (vp):

want to know more information about a specific person

Stalker (n):

a person who pursues someone obsessively and aggressively to the point of harassment

Stranger (n):

someone you do not know

Suspicious (adj):

doubtful; skeptical; wary.

Sustainability (n):

the quality of being able to continue over a period of time

SUV (n):

sport utility vehicle.

Swap (n):

an act, instance, or process of exchanging one thing for another

T

Tempting (adj):

appealing to

Threat (n):

intimidation; menace; risk.

Throw one off:

to make one suddenly surprised or confused

Throw you off (v):

to make you suddenly surprised or confused

Top-up (v):

to make up to the full quantity, capacity, or amount

Traditional energy sources (np):

things used to create energy with environmental impact

Trust (v):

to believe that someone is good and honest and won't harm you

U

Unsaturated fat (n):

beneficial fats because they can improve blood cholesterol levels, ease inflammation, stabilize heart rhythms, and play a number of other beneficial roles.

Urge (v):

persistently to persuade (someone) to do something

V

Vass majority (np):

great majority

Vegan (n):

a person who does not eat any food derived from animals and who typically does not use other animal products.

Vegan diet:

avoid all animal products including meat, egg and dairy

Violence (n):

rampage; assault; force.

W

Waste disposal (np):

a device in a kitchen sink that grinds up food waste so it can be washed down the drain

Windmill (n):

a mill or machine operated by the wind

Withdraw (v):

remove or take away (something) from a particular place or position.